MSG No. 128.400 $P4/mnc1'$ [Type II, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4/m..1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,6,9,10,11,14,17,18,19,22,25,26,27,30]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,5,7,8,12,13,15,16,20,21,23,24,28,29,31,32]`

Table 2: Wyckoff site: `2b`, site symmetry: `4/m..1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,6,9,10,11,14,17,18,19,22,25,26,27,30]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,5,7,8,12,13,15,16,20,21,23,24,28,29,31,32]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m..1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,6,9,14,17,22,25,30]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,3,10,11,18,19,26,27]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,5,12,13,20,21,28,29]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[7,8,15,16,23,24,31,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.221'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,6,7,8,17,22,23,24]`
2	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[2,3,4,5,18,19,20,21]`
3	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9,14,15,16,25,30,31,32]`
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[10,11,12,13,26,27,28,29]`

Table 5: Wyckoff site: `4e`, site symmetry: `4..1'`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,6,17,18,19,22]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[4,5,7,8,20,21,23,24]`
3	$[0, 0, -z]$	`[9,10,11,14,25,26,27,30]`
4	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[12,13,15,16,28,29,31,32]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,6,17,22]`
2	$\left[\frac{1}{2}, 0, z\right]$	`[2,3,18,19]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	`[4,5,20,21]`

continued ...

Table 6

No.	position	mapping
4	$\left[0,\, \frac{1}{2},\, \frac{1}{2}-z\right]$	[7,8,23,24]
5	$\left[0,\, \frac{1}{2},\, -z\right]$	[9,14,25,30]
6	$\left[\frac{1}{2},\, 0,\, -z\right]$	[10,11,26,27]
7	$\left[\frac{1}{2},\, 0,\, z+\frac{1}{2}\right]$	[12,13,28,29]
8	$\left[0,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[15,16,31,32]

Table 7: Wyckoff site: 8g, site symmetry: ..21'

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$	[1,7,17,23]
2	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{4}\right]$	[2,5,18,21]
3	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{4}\right]$	[3,4,19,20]
4	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	[6,8,22,24]
5	$\left[-x,\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	[9,15,25,31]
6	$\left[x+\frac{1}{2},\, -x,\, \frac{3}{4}\right]$	[10,13,26,29]
7	$\left[\frac{1}{2}-x,\, x,\, \frac{3}{4}\right]$	[11,12,27,28]
8	$\left[x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$	[14,16,30,32]

Table 8: Wyckoff site: 8h, site symmetry: m..1'

No.	position	mapping
1	$[x,\, y,\, 0]$	[1,14,17,30]
2	$[-y,\, x,\, 0]$	[2,11,18,27]
3	$[y,\, -x,\, 0]$	[3,10,19,26]
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[4,13,20,29]
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[5,12,21,28]
6	$[-x,\, -y,\, 0]$	[6,9,22,25]
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[7,16,23,32]
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[8,15,24,31]

Table 9: Wyckoff site: 16i, site symmetry: 11'

No.	position	mapping
1	$[x,\, y,\, z]$	[1,17]
2	$[-y,\, x,\, z]$	[2,18]
3	$[y,\, -x,\, z]$	[3,19]
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[4,20]
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[5,21]
6	$[-x,\, -y,\, z]$	[6,22]
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[7,23]
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[8,24]
9	$[-x,\, -y,\, -z]$	[9,25]
10	$[y,\, -x,\, -z]$	[10,26]

continued ...

Table 9

No.	position	mapping
11	$[-y,\, x,\, -z]$	`[11,27]`
12	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[12,28]`
13	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[13,29]`
14	$[x,\, y,\, -z]$	`[14,30]`
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[15,31]`
16	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[16,32]`